VINCENTIUS

Segitiga dan Lingkaran

$$s = \frac{a+b+c}{2} \qquad r = \frac{L_\Delta}{s}$$

$$L_\Delta = \sqrt{s(s-a)(s-b)(s-c)} \qquad R = \frac{abc}{4L_\Delta}$$

$$U_n = n \leftrightarrow S_n = \frac{1}{2}n(n+1)$$

$$U_n = n^2 \leftrightarrow S_n = \frac{1}{6}n(n+1)(2n+1)$$

$$U_n = n^3 \leftrightarrow S_n = \left(\frac{1}{2}n(n+1)\right)^2$$

$$U_n = n(n+1) \leftrightarrow S_n = \frac{1}{3}n(n+1)(n+2)$$

$$U_n = n(n+1)(n+2) \leftrightarrow S_n = \frac{1}{4}n(n+1)(n+2)(n+3)$$

$$U_n = \frac{1}{2}n(n+1) \leftrightarrow S_n = \frac{1}{6}n(n+1)(n+2)$$

Cermin: $\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$

Gusuran/Sharing: $\begin{bmatrix} 1 & k_x \\ k_y & 1 \end{bmatrix}$

Rotasi: $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

Regangan: $\begin{bmatrix} k_x & 0 \\ 0 & k_y \end{bmatrix}$

Dilatasi: $\begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$

AVM

VINCENTIUS

Sebaran Binom: $P_{(x,n)} = C_{(n,x)}(p)^x(q)^{(n-x)}$

Sebaran Hipergeometrik: $\mathrm{P(x, N, n, k)} = \frac{\mathrm{C_{(k,x)}C_{(N-k,n-x)}}}{\mathrm{C_{(N,n)}}}$

n pengambilan, diminta ***x***, ***N*** populasi, ***k*** unsur sukses

Pembagian Istimewa

$$\frac{\mathrm{a^n - b^n}}{\mathrm{a-b}} = \mathrm{a^{n-1} + a^{n-2}b + \cdots + ab^{n-2} + b^{n-1}}$$

$$\frac{\mathrm{a^{2n} - b^{2n}}}{\mathrm{a+b}} = \mathrm{a^{2n-1} - a^{2n-2}b + \cdots + ab^{2n-2} - b^{2n-1}}$$

$$\frac{\mathrm{a^{2n+1} + b^{2n+1}}}{\mathrm{a+b}} = \mathrm{a^{2n} - a^{2n-1}b + \cdots - ab^{2n-1} + b^{2n}}$$